Deep End-to-End Causal Inference: Bayesian Structure Learning and Simulation-Based Treatment Effect Estimation under Heterogeneous, Partially Observed Data

CSE 516 – Probabilistic Graphical Models
Major Project – M4 Final Report
Track: Theoretical AI
Group: G5_TH_PGM

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Repository: https://github.com/masoomchoksi/G5_TH_PGM (folder: M4_G5_TH_PGM)

Abstract

Causal inference asks how an intervention on one variable changes another, but the true causal graph is usually unknown. Deep End-to-End Causal Inference (DECI) combines Bayesian causal-graph learning, structural equation modelling and simulation-based effect estimation in one probabilistic framework. This project investigates the graph-recovery guarantee (Theorem 1) of DECI: the assumptions needed for recovering the true graph, and what happens when they are relaxed, with focus on categorical/mixed-type variables and partially observed data.

First, we establish the research problem, datasets, and baselines, and reconstruct the main steps of Theorem 1. We then implement DECI-G and DECI-S on Microsoft Causica—comparing them against NOTEARS-MLP and ICA-LiNGAM—and propose a two-stage Graph-Aware Imputation method based on Markov Blankets to handle missing data.

Our results show that DECI's performance depends heavily on the underlying data-generating process, that strong adjacency recovery does not guarantee accurate edge orientation (e.g., on linear exponential data, DECI-G achieves an adjacency F1 of 1.000 but an orientation F1 of 0.013), and that recovery degrades as graph complexity increases. While we report baseline DECI-G results under missingness, comparative evaluation of Graph-Aware Imputation remains ongoing; accordingly, we make no claims of improved imputation performance, new identifiability guarantees, or overall superiority over existing baselines.

1. Introduction and Motivation

Causal questions ("what happens to exam marks if study hours are actively increased?") require a causal graph, usually a DAG whose nodes are variables and whose directed edges are direct causal relations. Classical pipelines first discover a graph and then estimate effects, so graph errors propagate. DECI (Geffner et al., 2022) instead learns a distribution over graphs jointly with structural mechanisms and estimates effects by simulating interventions.

A model can perform well empirically without satisfying the assumptions behind its theoretical guarantee. This project therefore separates four things: recovery of the observational distribution, identification of the true graph, practical convergence of training, and accuracy of downstream causal-effect estimates.

Research questions

1. Can the argument of DECI Theorem 1 be reconstructed from its stated assumptions?
2. How does maximum-likelihood recovery of the joint distribution relate to graph recovery?
3. When does ELBO optimization approach the maximum-likelihood solution?
4. Why can categorical variables create ambiguity in edge orientation?
5. Why does approximate amortised imputation of missing values fall outside the original proof?
6. How do these limitations appear in graph recovery and ATE/CATE estimation?

2. Problem Formulation

Causal graph. $G = (V, E)$ is a DAG; an edge $X_i \rightarrow X_j$ means X_i is a direct cause of X_j. The graph is unknown and estimated from observational data.

Structural equation model. Nonlinear additive-noise model: $X_i = f_i(X_{pa(i)}) + Z_i$, with structural function f_i and exogenous noise Z_i.

Bayesian structure learning. DECI learns a posterior $q_\varphi(G)$ over graphs together with mechanism and noise parameters.

Intervention. $do(X_T = a)$ replaces the mechanism of the treatment X_T by the value a; learned equations are used to simulate outcomes.

ATE / CATE. ATE compares $E[Y \mid do(X_T = a)]$ with $E[Y \mid do(X_T = b)]$; CATE conditions on covariates of a subgroup. DECI estimates both by simulation over structural equations and graph samples.

3. Background

PGM view. Representation (nodes, directed edges, DAG), Learning (graph posterior and mechanism parameters), Inference (intervention simulation).

Identifiability. Different DAGs can produce the same observational distribution (Markov equivalence). Linear Gaussian models are generally not identifiable from observational data alone; linear non-Gaussian

and suitable nonlinear additive-noise models can be. Recovering the distribution therefore does not by itself recover the graph.

Variational inference. DECI approximates the graph posterior with $q_\varphi(G)$ and maximises the ELBO, which balances data fit against KL regularisation toward the graph prior.

4. Related Work

- **DECI** (Geffner et al., 2022): joint graph and mechanism learning with simulation-based effect estimation; our principal model.
- **NOTEARS** (Zheng et al., 2018): continuous optimisation with a differentiable acyclicity constraint; NOTEARS-MLP is our nonlinear baseline.
- **PC**: constraint-based discovery using conditional-independence tests.
- **ICA-LiNGAM**: linear non-Gaussian discovery via independent component analysis.
- **Additive-noise models** (Peters et al., 2014): identifiability results that underpin DECI's recovery argument.
- **GraN-DAG**: neural nonlinear DAG learning (listed as a considered method).

5. State-of-the-Art Positioning

Anchor (selected in M1): Vashishtha et al., "Causal Order: The Key to Leveraging Imperfect Experts in Causal Inference," ICLR 2025. It studies how imperfect expert (human or LLM) judgments can yield causal-order information; pairwise judgments can confuse direct and indirect relations and create cycles, and the paper uses causal-order and triplet-style prompting for more cycle-consistent information.

Relationship. That work uses expert-derived information; DECI learns a graph distribution from observational data. Our project analyses DECI's recovery assumptions and their limits. The comparison is analytical and complementary, not a numerical benchmark.

Final position. Where we are stronger: a transparent reconstruction of the Theorem 1 argument, explicit assumption audit, and identification of categorical and missing-data boundaries. Where we fall short: comparative M3 results for the proposed method are incomplete, and no direct numerical comparison with the SOTA paper exists.

5.4 Side-by-side comparison with the SOTA anchor

Dimension	Vashishtha et al. (ICLR 2025)	This project
Information source	Imperfect expert judgments (human or LLM) about causal order	Observational data, fully or partially observed

Core problem	Pairwise judgments confuse direct and indirect relations and can form cycles	When DECI's graph-recovery guarantee holds and where it breaks
Approach	Causal-order information and triplet-style prompting for cycle-consistent output	Theorem 1 reconstruction, assumption audit, categorical and missing-data boundaries, proposed Graph-Aware Imputation
Evaluation	Cycle formation, causal-order accuracy, downstream causal tasks	SHD, adjacency/orientation F1, ATE RMSE, imputation RMSE on selected settings
Key reported numbers	Child graph, SHD: GPT-4 pairwise 148 (>10,000 cycles) vs triplet with Phi-3 69 and Llama-3 8B 129 (0 cycles) (Tables 2, A7, A8). Asia-M, human annotators: pairwise 3 cycles, triplet 0 (Table 1). Asia, causal-effect error: CaMML 0.05 vs LLM order + CaMML 0.00 (Table 5).	Section 12
Comparable on a shared benchmark?	No	No

What the anchor achieves. It shows that asking experts for causal order, rather than for isolated pairwise edges, yields more cycle-consistent causal information. On the 20-node Child graph, triplet prompting with the small models Phi-3 and Llama-3 8B (GPT-4 used for tie-breaking) produced no cycles, while pairwise prompting with GPT-4 produced more than 10,000 (SHD 69 and 129 versus 148). With human annotators on Asia-M, pairwise prompting produced 3 cycles and triplet prompting none. Downstream, giving the LLM-derived order to CaMML reduced the average-causal-effect error on Asia from 0.05 to 0.00 over 3 runs. These metrics (SHD, cycle counts, D_top, causal-effect error) differ from ours, so the numbers give context and are not a like-for-like comparison.

Where this project is strong. It makes DECI's guarantee explicit (assumptions A1–A10), separates adjacency from orientation recovery, and shows where the proof does not transfer to categorical and missing-data settings.

Where it falls short. The proposed Graph-Aware Imputation has no completed comparison, several planned experiments have no reported results, and there is no shared-benchmark comparison with the anchor.

Possible link (untested idea, not a claim). Causal-order information of the kind the anchor studies could in principle inform a graph prior for DECI. We have not tested this.

6. Graphical Representation

- **Nodes/edges:** each variable is a node; X_j → X_i means X_j is a parent of X_i; the adjacency matrix encodes the graph.
- **Structural functions:** neural mechanisms using residual MLP blocks, shared weights and node embeddings.
- **Noise:** Gaussian and rational-quadratic spline configurations; independence of noise matters for identifiability.
- **Graph prior:** sparsity term plus acyclicity penalty h(G) = tr(exp(G◦G)) − D (D = number of variables), non-negative and zero for DAGs; the penalty weight is adjusted by an augmented-Lagrangian schedule.
- **Graph posterior:** Bernoulli edge variables with the ENCO parameterisation (existence and orientation modelled separately). The mean-field form cannot capture all dependencies between edges.

6.6 Formal specification of the model

Note: the equations below are the standard DECI formulation. Check the symbols against the notation in your M2 report before submission.

Let A ∈ {0,1}^{D×D} be the adjacency matrix of G. Each variable has a mechanism with additive noise:

$$\mathbf{x}_i = f_i\left(x_{\mathrm{pa}_G(i)}\right) + z_i, \qquad z_i \sim p_{z_i}$$

For a DAG the joint density factorises over parents:

$$\mathrm{p}_\theta(x \mid G) = \prod_{i=1}^{D} p_{z_i}\left(x_i - f_i(x_{\mathrm{pa}_G(i)})\right)$$

The graph prior combines sparsity with the acyclicity penalty, where ρ and α are the augmented-Lagrangian parameters:

$$\mathrm{p(G)} \propto \exp\Big(-\lambda_s A_F^2 - \alpha\, h(A) - \rho 2\, h(A)^2 \Big), \qquad h(A) = tr\big(e^{A \circ A}\big) - D$$

The variational graph posterior is a product of independent Bernoulli edges (mean-field). With the ENCO parameterisation, edge existence and orientation have separate logits γ and θ:

$$q_\phi(G) = \prod_{i \neq j} q_\phi(A_{ij}), \qquad q_\phi(A_{ij} = 1) = \sigma(\gamma_{ij})\,\sigma(\theta_{ij}), \quad q_\phi(A_{ji} = 1) = \sigma(\gamma_{ij})\,\sigma(-\theta_{ij})$$

Gradients through the discrete edge samples use the Gumbel-softmax relaxation (temperature 0.25 in the reference configuration). Because q factorises over edges, it cannot represent dependencies between edges, which is the limitation noted in Section 15.

7. Learning Method

DECI jointly learns the graph posterior, structural-function parameters and noise parameters by maximising the ELBO. The normalised objective is (1/N)·E_q[log p_θ(x | G)] − (1/N)·KL(q_φ ‖ p). The first term measures data fit, the second regularises toward the prior.

Reference configuration (from M2):

Component	Setting
Structural MLP	2 hidden layers, 128 units, residual connections, layer normalisation
Noise model	Rational-quadratic spline, 8 bins
Sparsity parameter	λ_s = 5
Augmented Lagrangian (initial)	ρ = 1, α = 0
Optimiser	Adam, learning rate 0.01

Gradient estimator	Gumbel-softmax, temperature 0.25

Software versions and hardware:

- `causica==0.4.6` (Microsoft Causica framework for DECI)
- `torch==2.14.1` (PyTorch)
- `pytorch-lightning==2.6.6` (Training orchestration)
- `tensordict==0.15.6` (Tensor data structures used by Causica)
- `torchmetrics==1.9.0` (Evaluation metrics)

8. Inference Method

- **Graph posterior:** candidate graphs are sampled from q_φ(G) rather than relying on a single point estimate.
- **ATE:** sample a graph, sample noise, apply the intervention, simulate the structural equations, average over simulations and graph samples. Reference configuration: 1,000 graph samples, 2 noise realisations.
- **CATE:** surrogate regression on simulated intervention outcomes using a random-Fourier-feature linear surrogate; configuration: 10 graph samples, 10,000 realisations, 3,000 random Fourier features.
- **Trade-off:** exact inference over all graphs is expensive and nonlinear flows rule out closed-form message passing, so simulation is used, at the cost of Monte Carlo variability and dependence on posterior quality.

8.5 Formal inference procedure

Note: standard formulation; verify against your M2 implementation notes.

An intervention do(X_T = a) on a sampled graph G_s removes all edges into X_T and fixes its value. The remaining variables are simulated in the topological order of the mutilated graph.

Algorithm (ATE). For s = 1..S: sample G_s from q_φ(G). For k = 1..K: sample noise z^{(s,k)} from the learned noise model, set x_T = a, and compute each other x_i in topological order as f_i(x_pa) + z_i. Record the outcome y^{(s,k)}(a). Repeat for b.

$$\widehat{E}[Y \mid do(X_T = a)] = \frac{1}{SK} \sum_{s=1}^{S} \sum_{k=1}^{K} y^{(s,k)}(a), \qquad \widehat{\mathrm{ATE}} = \widehat{E}[Y \mid do(X_T = a)] - \widehat{E}[Y \mid do(X_T = b)]$$

The reference configuration uses S = 1,000 graph samples and K = 2 noise realisations.

CATE by surrogate regression. Simulated intervention outcomes are regressed on the conditioning covariates with a linear model on random Fourier features φ(x):

$$\hat{w} = \arg\min_w \sum_m \left(y_m - w^\top \phi(x_m)\right)^2 + \lambda w^2, \qquad \widehat{\mathrm{CATE}}(x) = \hat{w}_a^\top \phi(x) - \hat{w}_b^\top \phi(x)$$

The reference configuration uses 10 graph samples, 10,000 realisations and 3,000 random Fourier features. The ridge penalty λ is 5.

Sources of error. (i) Monte Carlo noise from finite S and K; (ii) error in the learned graph posterior; (iii) mechanism misspecification; (iv) surrogate approximation error for CATE. Each cost per estimate scales with S × K forward passes through D mechanisms.

9. Theoretical Analysis (Theorem 1)

Assumptions. A1 minimality; A2 structural identifiability; A3 correct specification; A4 causal sufficiency; A5 regularity (finite expected log-likelihoods). Implicit conditions identified in our audit: A6 uniform-convergence/argmax conditions; A7 representability of the limiting optimum by the variational family; A8 graph support and cyclic-graph probability; A9 global optimum vs practical optimisation; A10 bounded prior hyperparameters.

Argument in five steps.

1. Valid DAG likelihood: topological ordering gives a triangular Jacobian, supporting the likelihood. This does not automatically cover cyclic samples, so the cyclic-sample rate is kept as an experimental diagnostic.
2. Prior and acyclicity: h(G) is finite, non-negative, and zero for DAGs.
3. Maximum-likelihood recovery: by Gibbs' inequality the expected log-likelihood is maximised when the model distribution equals the true one under correct specification. This recovers the distribution, not necessarily a unique graph.
4. Structural identifiability: under A2, the distribution corresponds to a unique graph in the model class (e.g. nonlinear continuous additive-noise models under suitable conditions).
5. ELBO convergence: under bounded hyperparameters and prior assumptions, KL(q_φ ‖ p) = O(1), hence (1/N)·KL → 0 as N → ∞. Convergence of the objective alone does not give convergence of the optimiser; A6–A9 are also needed, and Adam gives no global-optimality guarantee.

Boundary 1 – categorical variables (proposition, not a theorem). For two binary variables, p(x1, x2) = p(x1)p(x2 | x1) = p(x2)p(x1 | x2), so X1 → X2 and X2 → X1 can explain the same distribution. Structural identifiability may fail and the posterior may concentrate on an equivalence class. Prediction: adjacency may be recoverable while direction stays uncertain, so adjacency and orientation must be evaluated separately. This requires empirical verification.

Boundary 2 – missing data. Missing entries become per-observation latent variables, so the variational gap may scale with N and the vanishing-normalised-KL argument cannot be transferred directly. A Gaussian graph-agnostic imputer may mismatch the true nonlinear/non-Gaussian conditional. No missing-data recovery theorem is established.

Status separation. Proven/argued in reports: Steps 1–5 under stated assumptions. Proposed: categorical equivalence-class conclusion, Graph-Aware Imputation. Experimental observations: Section 12.

9.1 Fuller derivations

Note: these are standard derivations written out in full. Cross-check them with the proof analysis in your M2 and M3 reports and keep whichever notation those use.

Lemma 1 (valid DAG likelihood). Write the model as z = x − F_G(x). For a DAG, order the variables topologically; then ∂F_G/∂x is strictly lower-triangular, so the Jacobian of the map x ↦ z is unit lower-triangular and has determinant 1. By change of variables:

$$p_\theta(x \mid G) = p_z\big(x - F_G(x)\big)\left|\det\left(I - \partial F_G/\partial x\right)\right| = \prod_i p_{z_i}\left(x_i - f_i(x_{\mathrm{pa}(i)})\right)$$

For a cyclic graph no such ordering exists and the determinant need not be 1, so the density formula is not justified. This is why the cyclic-sample rate is tracked as a diagnostic.

Lemma 2 (Gibbs' inequality, distribution recovery). Let p* be the true data distribution. Then

$$\mathrm{E}_{p^*}[\log p_\theta(x)] - E_{p^*}[\log p^*(x)] = -\mathrm{KL}(p^* \| p_\theta) \le 0$$

with equality only if p_θ = p* almost everywhere. Under A3 (correct specification) the maximum is attained, so the maximiser reproduces the true distribution. By itself this does not select a graph.

Lemma 3 (identifiability closes the gap). If A2 holds, the only pair (G, θ) in the model class with p_{G,θ} = p* is the true one. Combined with Lemma 2, the likelihood maximiser has the true graph.

Lemma 4 (ELBO decomposition). For any q_φ(G):

$$\mathrm{ELBO}(\theta,\phi) = E_{q_\phi}\left[\log p_\theta(X \mid G)\right] - \mathrm{KL}\big(q_\phi \| p\big) = \log p_\theta(X) - \mathrm{KL}\big(q_\phi \| p(G \mid X)\big) \le \log p_\theta(X)$$

Lemma 5 (KL is O(1)). The graph space is finite (at most 2^{D(D-1)} directed graphs). If the prior has p(G) ≥ ε > 0 on the support of q for bounded hyperparameters (A10), then since −H(q) ≤ 0:

$$\mathrm{KL}(q \| p) = -H(q) - \textstyle\sum_G q(G)\log p(G) \;\le\; \log(1/\varepsilon)$$

The bound does not depend on N, so (1/N)·KL → 0.

Proposition 1 (what follows for the optimiser). Dividing the ELBO by N gives the average log-likelihood term minus a vanishing term. If the average log-likelihood converges uniformly in (θ, q) and the limiting optimum is unique and representable by the variational family (A6, A7), the maximisers converge to the likelihood maximiser, and by Lemma 3 to the true graph. This is a conditional statement. It does not claim that Adam reaches that optimum (A9).

Proposition 2 (categorical boundary). For binary X1, X2 the joint has 3 free parameters. Both factorisations p(x1)p(x2|x1) and p(x2)p(x1|x2) have 3 free parameters and can represent every joint exactly, so both attain the same maximum likelihood. Structural identifiability (A2) fails, and with equal sparsity cost the prior does not break the tie, so posterior mass is expected to split across the two directions. This is a proposition from the analysis, not an empirically verified result.

Proposition 3 (missing data breaks Lemma 5). With missing entries, each observation n has its own latent value z_n and its own approximate posterior q_n. The total gap is a sum over observations:

$$\sum_{n=1}^{N} \mathrm{KL}\big(q_n(z_n) \| p_\theta(z_n \mid x_n^{\mathrm{obs}}, G)\big)$$

If the amortised family cannot match the true conditional, each term is bounded below by some $\delta > 0$, so the sum grows like $N\delta$ and $(1/N)\cdot(\text{gap})$ does not vanish. The normalised-KL argument of Lemma 5 therefore does not carry over. Graph-Aware Imputation changes the family of q_n, but being a preprocessing step it leaves this structure of the objective unchanged.

10. Proposed Method: Graph-Aware Imputation (M3)

Instead of conditioning on all other variables, the imputer conditions on the **Markov Blanket** of the variable with missing values, as a structural inductive bias. It adds no new information to the data and its benefit is not established.

Variant	Description	Role
A	Original DECI (graph-agnostic amortised imputation)	Required control
B	Mean/median fill, then refit	Basic control
C	Agnostic MLP using all other variables	Separates structure from imputer capacity

D	Graph-aware deterministic (MLP on Markov Blanket)	First prototype
E	Graph-aware stochastic (predicts distribution parameters, samples)	Preferred proposal
F	Oracle true graph	Diagnostic only, not a fair method

Pipeline: (1) synthetic data with known DAG; (2) apply missingness; (3) run DECI on incomplete data; (4) sample graphs from the posterior; (5) find Markov Blankets of variables with missing values; (6) impute from blanket variables; (7) repeat when blanket variables are also missing; (8) preserve observed entries; (9) retrain DECI on completed data; (10) evaluate graph recovery, imputation quality and causal effects. Missingness levels 0%, 10%, 30%, 50%; the same mask should be used across methods for each seed and rate.

Limitation: this is a two-stage preprocess-and-refit procedure; it does not modify the ELBO, so the imputation gap, Gaussian-family mismatch and observed-data identifiability remain open.

11. Experimental Setup, Baselines and KPIs

Planned datasets (M1): CSuite; ER/scale-free random DAGs; SynTReN; Sachs; IHDP; Twins. Being planned does not mean every experiment was completed.

Methods: DECI-G, DECI-S, NOTEARS-MLP, PC, ICA-LiNGAM; DoWhy linear/nonlinear DML pipelines were identified in M1 for compatible effect comparisons.

Protocol (as specified): matched data and splits, documented seeds, multiple runs, consistent conditions, recorded environment, missing runs reported rather than dropped.

KPIs

Stage	Metric
Representation	SHD (lower is better), adjacency F1, orientation F1, posterior mass on true graph, cyclic-sample rate, ordering accuracy

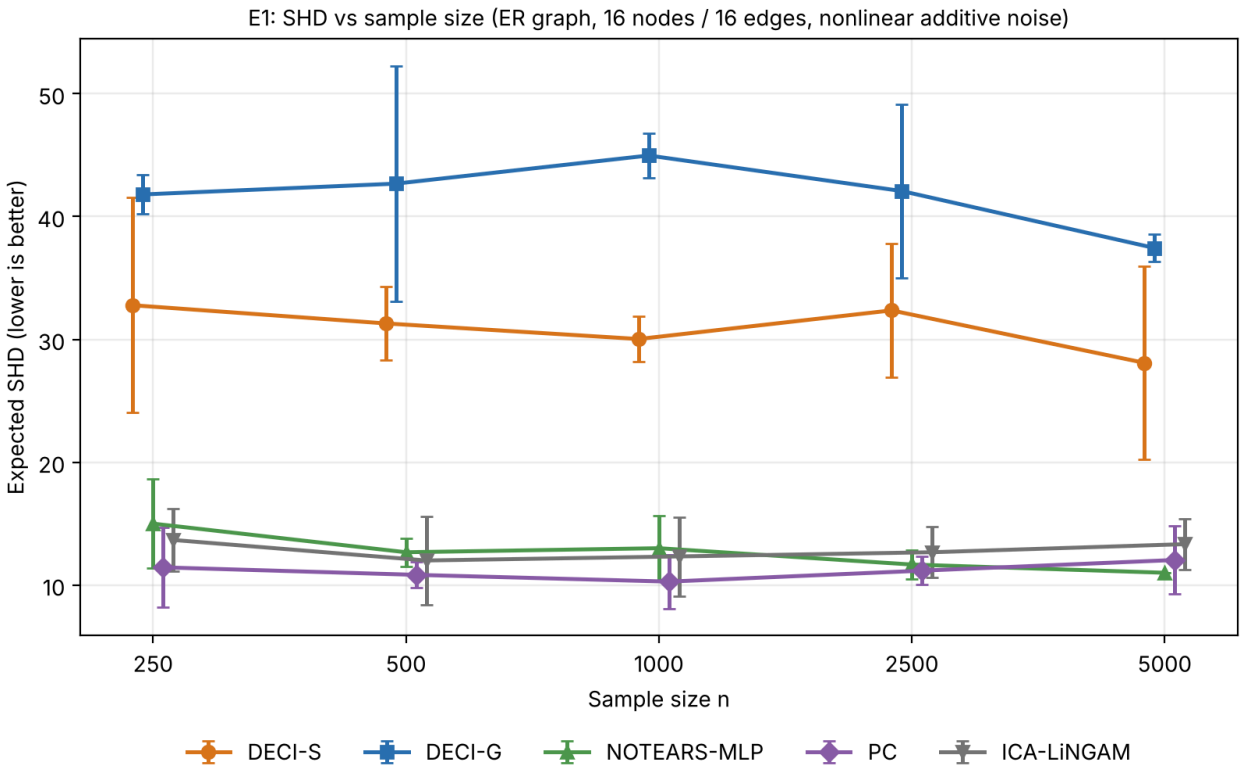
Learning	ELBO, held-out NLL, log-loss
Inference	ATE RMSE, CATE RMSE, runtime, Monte Carlo variability
Missing data	Imputation RMSE and MAE, held-out NLL, KL to known conditional where computable
Probability quality	NLL, log-loss, Brier score, ECE

12. Results

12.1 E1 sample-size experiment

Synthetic Erdős–Rényi graph, 16 nodes, 16 edges, nonlinear additive-noise model with spline-warped noise. Metric: expected SHD (mean ± SD; lower is better).

n	DECI-S	DECI-G	NOTEARS-MLP	PC	ICA-LiNGAM
250	32.78 ± 8.75	41.80 ± 1.58	15.00 ± 3.61	11.44 ± 3.22	13.67 ± 2.52
500	31.30 ± 3.00	42.68 ± 9.60	12.67 ± 1.15	10.83 ± 1.04	12.00 ± 3.61
1000	30.03 ± 1.82	44.97 ± 1.83	13.00 ± 2.65	10.28 ± 2.18	12.33 ± 3.21
2500	32.36 ± 5.44	42.08 ± 7.07	11.67 ± 1.15	11.17 ± 1.15	12.67 ± 2.08
5000	28.09 ± 7.89	37.44 ± 1.12	11.00 ± 0.00	12.04 ± 2.77	13.33 ± 2.08



DECI variants improve somewhat at larger n but not consistently; PC and NOTEARS-MLP generally have lower SHD here. The prediction that more data consistently improves DECI is only partially supported.

12.2

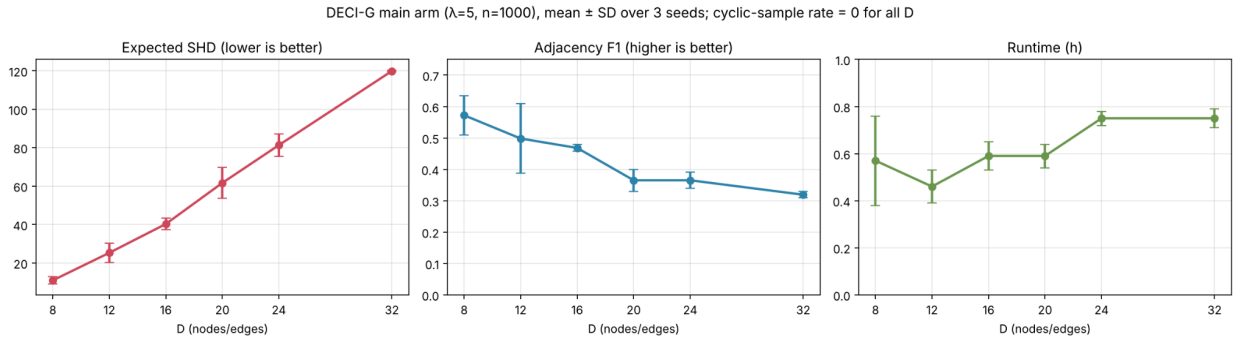
E2 (a) DECI-G main arm (λ = 5, n = 1000), mean ± SD over 3 seeds

Setting	Expected SHD	Adj. F1	Cyclic-sample rate	Median shortest cycle	Runtime (h)
D = 8, 8 edges	10.91 ± 1.97	0.572 ± 0.062	0.000 ± 0.000	none cyclic	0.57 ± 0.19
D = 12, 12 edges	25.25 ± 4.96	0.498 ± 0.111	0.000 ± 0.000	none cyclic	0.46 ± 0.07
D = 16, 16 edges	40.26 ± 3.01	0.468 ± 0.011	0.000 ± 0.000	none cyclic	0.59 ± 0.06

D = 20, 20 edges	61.65 ± 7.93	0.365 ± 0.035	0.000 ± 0.000	none cyclic	0.59 ± 0.05
D = 24, 24 edges	81.30 ± 5.81	0.365 ± 0.026	0.000 ± 0.000	none cyclic	0.75 ± 0.03
D = 32, 32 edges	119.89 ± 0.45	0.319 ± 0.010	0.000 ± 0.000	none cyclic	0.75 ± 0.04

E2 (b) DECI-S validation arm (λ = 5, n = 1000), mean ± SD over 2 seeds

Setting	Expected SHD	Adj. F1	Cyclic-sample rate	Median shortest cycle	Runtime (h)
D = 8, 8 edges	9.67 ± 2.09	0.663 ± 0.082	0.005 ± 0.007	3	1.58 ± 0.02
D = 24, 24 edges	69.78 ± 0.77	0.365 ± 0.039	0.000 ± 0.000	none cyclic	4.84 ± 0.70

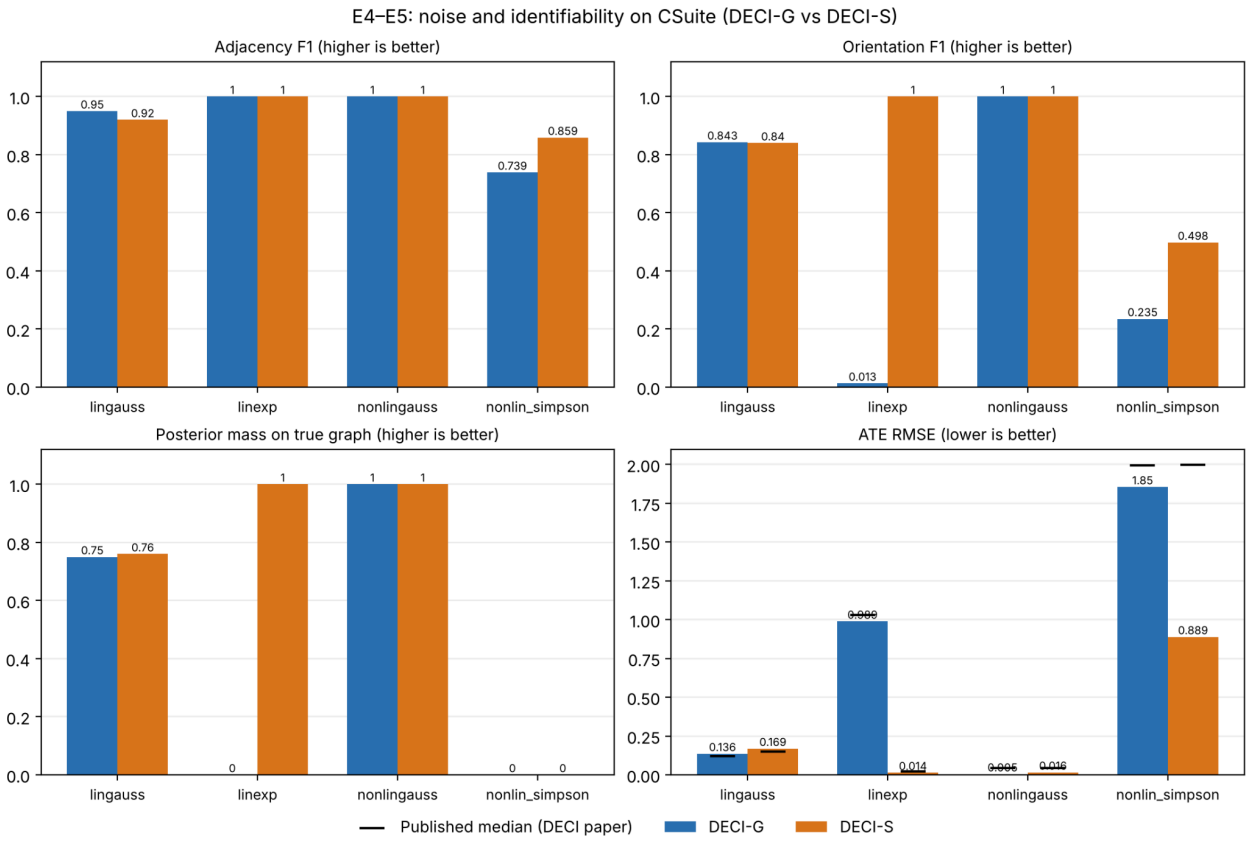


12.3 E4 -E5 noise and identifiability (CSuite)

Dataset	Method	Adj. F1	Orient. F1	Posterior mass on true graph	ATE RMSE (ours)	Published median ATE RMSE

lingauss	DECI-G	0.950	0.843	0.750	0.136	0.120
lingauss	DECI-S	0.920	0.840	0.760	0.169	0.149
linexp	DECI-G	1.000	0.013	0.000	0.989	1.029
linexp	DECI-S	1.000	1.000	1.000	0.014	0.022
nonlingauss	DECI-G	1.000	1.000	1.000	0.005	0.042
nonlingauss	DECI-S	1.000	1.000	1.000	0.016	0.043
nonlin_simpson	DECI-G	0.739	0.235	0.000	1.855	1.995
nonlin_simpson	DECI-S	0.859	0.498	0.000	0.889	1.997

The published-median column is a reference from the original DECI paper, not a result produced by this project. In linexp both variants find the skeleton (adjacency F1 1.000) but only DECI-S gets orientation right, showing that adjacency recovery does not imply orientation recovery. Neither variant is uniformly best.



12.4 E7 missingness (DECI-G baseline)

Nonlinear

Missing	Expected SHD	Adj. F1	Orient. F1	Min. SHD (edges − 16)	Imputation RMSE	Mean-fill RMSE	Runtime (h)	ATE RMSE	CATE RMSE
0%	40.26 ± 3.01	0.468 ± 0.011	0.216 ± 0.031	≥ 29.2	—	—	0.54 ± 0.02	0.360 ± 0.109	0.423 ± 0.142
10%	47.36 ± 3.88	0.380 ± 0.026	0.126 ± 0.024	≥ 31.3	0.807 ± 0.011	1.000 ± 0.003	0.60 ± 0.04	0.390 ± 0.095	0.393 ± 0.095

30%	42.10 ± 3.93	0.429 ± 0.031	0.197 ± 0.014	≥ 29.2	0.816 ± 0.046	1.001 ± 0.013	0.46 ± 0.06	0.367 ± 0.052	0.456 ± 0.160
50%	36.87 ± 4.54	0.484 ± 0.066	0.237 ± 0.063	≥ 25.5	0.855 ± 0.036	0.995 ± 0.010	0.43 ± 0.03	0.375 ± 0.047	0.403 ± 0.070

Linear-Gaussian

Missing	Expected SHD	Adj. F1	Orient. F1	Min. SHD (edges − 16)	Imputation RMSE	Mean-fill RMSE	Runtime (h)	ATE RMSE	CATE RMSE
0 %	53.03 ± 2.76	0.337 ± 0.014	0.089 ± 0.029	≥ 35.4	—	—	0.48 ± 0.04	0.516 ± 0.039	0.461 ± 0.030
10 %	42.17 ± 7.14	0.449 ± 0.090	0.215 ± 0.060	≥ 30.8	0.643 ± 0.061	1.011 ± 0.013	0.66 ± 0.08	0.518 ± 0.054	0.465 ± 0.098
30 %	42.46 ± 6.49	0.442 ± 0.049	0.207 ± 0.063	≥ 30.6	0.657 ± 0.049	0.992 ± 0.019	0.52 ± 0.05	0.524 ± 0.067	0.479 ± 0.124
50 %	41.21 ± 3.78	0.474 ± 0.035	0.180 ± 0.059	≥ 29.1	0.725 ± 0.046	1.003 ± 0.010	0.43 ± 0.04	0.530 ± 0.016	0.462 ± 0.063

12.5 E7 missingness (DECI-S baseline)

Nonlinear

Missing	Expected SHD	Adj. F1	Orient. F1	Min. SHD (edges − 16)	Imputation RMSE	Mean-fill RMSE	Runtime (h)	ATE RMSE	CATE RMSE
0 %	33.10 ± 6.26	0.491 ± 0.083	0.316 ± 0.067	≥ 23.2	—	—	2.35 ± 0.53	0.341 ± 0.064	0.281 ± 0.042
10 %	28.28 ± 8.25	0.574 ± 0.100	0.345 ± 0.146	≥ 19.7	0.763 ± 0.033	1.000 ± 0.003	2.52 ± 0.20	0.349 ± 0.052	0.345 ± 0.105
30 %	28.54 ± 0.90	0.515 ± 0.011	0.384 ± 0.040	≥ 19.8	0.804 ± 0.035	1.001 ± 0.013	2.83 ± 0.83	0.295 ± 0.033	0.286 ± 0.037
50 %	29.29 ± 3.05	0.535 ± 0.045	0.400 ± 0.071	≥ 22.9	0.852 ± 0.030	0.995 ± 0.010	2.29 ± 0.57	0.309 ± 0.103	0.310 ± 0.040

Linear-Gaussian

Missing	Expected SHD	Adj. F1	Orient. F1	Min. SHD (edges − 16)	Imputation RMSE	Mean-fill RMSE	Runtime (h)	ATE RMSE	CATE RMSE
0 %	38.18 ± 3.60	0.478 ± 0.004	0.189 ± 0.063	≥ 25.2	—	—	2.28 ± 0.41	0.529 ± 0.062	0.406 ± 0.101
10 %	36.72 ± 4.27	0.438 ± 0.022	0.232 ± 0.019	≥ 23.1	0.615 ± 0.045	1.011 ± 0.013	2.59 ± 0.25	0.466 ± 0.040	0.397 ± 0.021

30 %	40.54 ± 7.09	0.466 ± 0.052	0.229 ± 0.077	≥ 29.9	0.651 ± 0.049	0.992 ± 0.019	2.29 ± 0.71	0.483 ± 0.051	0.384 ± 0.079
50 %	37.84 ± 4.69	0.437 ± 0.070	0.304 ± 0.073	≥ 27.9	0.734 ± 0.052	1.003 ± 0.010	1.93 ± 0.26	0.440 ± 0.051	0.425 ± 0.061

12.6 Graph-Aware Imputation with DECI-G (λ = 5, three seeds)

Nonlinear

Missing	Expected SHD	Adj. F1	Orient. F1	Min. SHD (edges − 16)	Imputation RMSE	Mean-fill RMSE	Runtime (h)	ATE RMSE	CATE RMSE
0 %	40.26 ± 3.01	0.468 ± 0.011	0.216 ± 0.031	≥ 29.2	—	—	0.54 ± 0.02	0.360 ± 0.109	0.423 ± 0.142
10 %	48.45 ± 3.49	0.408 ± 0.063	0.155 ± 0.078	≥ 35.4	0.810 ± 0.049	1.000 ± 0.003	1.17 ± 0.10	0.413 ± 0.112	0.410 ± 0.077
30 %	60.71 ± 2.94	0.339 ± 0.026	0.194 ± 0.025	≥ 50.7	0.863 ± 0.066	1.001 ± 0.013	1.43 ± 0.43	0.391 ± 0.103	0.394 ± 0.082
50 %	68.65 ± 4.92	0.266 ± 0.036	0.133 ± 0.007	≥ 53.7	0.938 ± 0.042	0.995 ± 0.010	1.13 ± 0.33	0.414 ± 0.089	0.433 ± 0.084

Linear-Gaussian

Missing	Expected SHD	Adj. F1	Orient. F1	Min. SHD (edges − 16)	Imputation RMSE	Mean-fill RMSE	Runtime (h)	ATE RMSE	CATE RMSE
0 %	53.03 ± 2.76	0.337 ± 0.014	0.089 ± 0.029	≥ 35.4	—	—	0.48 ± 0.04	0.516 ± 0.039	0.461 ± 0.030
10 %	52.31 ± 4.56	0.371 ± 0.044	0.099 ± 0.049	≥ 36.3	0.614 ± 0.049	1.011 ± 0.013	1.24 ± 0.13	0.515 ± 0.043	0.448 ± 0.064
30 %	48.99 ± 1.93	0.371 ± 0.080	0.113 ± 0.018	≥ 32.7	0.674 ± 0.057	0.992 ± 0.019	1.53 ± 0.44	0.534 ± 0.088	0.512 ± 0.103
50 %	65.57 ± 8.76	0.325 ± 0.034	0.125 ± 0.024	≥ 52.6	0.767 ± 0.062	1.003 ± 0.010	1.08 ± 0.31	0.489 ± 0.031	0.482 ± 0.082

12.7 Graph-Aware Imputation with DECI-S (λ = 5, three seeds)

Nonlinear

Missing	Expected SHD	Adj. F1	Orient. F1	Min. SHD (edges − 16)	Imputation RMSE	Mean-fill RMSE	Runtime (h)	ATE RMSE	CATE RMSE
0 %	33.10 ± 6.26	0.491 ± 0.083	0.316 ± 0.067	≥ 23.2	—	—	2.35 ± 0.53	0.341 ± 0.064	0.281 ± 0.042
10 %	33.20 ± 6.16	0.475 ± 0.063	0.329 ± 0.050	≥ 23.2	0.772 ± 0.043	1.000 ± 0.003	5.36 ± 0.28	0.313 ± 0.077	0.356 ± 0.069

30 %	45.14 ± 3.15	0.420 ± 0.067	0.194 ± 0.066	≥ 33.2	0.849 ± 0.055	1.001 ± 0.013	5.47 ± 0.86	0.399 ± 0.120	0.397 ± 0.114
50 %	54.35 ± 4.23	0.382 ± 0.026	0.207 ± 0.013	≥ 45.0	0.930 ± 0.047	0.995 ± 0.010	5.21 ± 0.90	0.389 ± 0.117	0.373 ± 0.060

Linear-Gaussian

Missing	Expected SHD	Adj. F1	Orient. F1	Min. SHD (edges − 16)	Imputation RMSE	Mean-fill RMSE	Runtime (h)	ATE RMSE	CATE RMSE
0 %	38.18 ± 3.60	0.478 ± 0.004	0.189 ± 0.063	≥ 25.2	—	—	2.28 ± 0.41	0.529 ± 0.062	0.406 ± 0.101
10 %	39.00 ± 4.20	0.457 ± 0.055	0.217 ± 0.023	≥ 26.7	0.616 ± 0.063	1.011 ± 0.013	5.22 ± 0.56	0.488 ± 0.063	0.394 ± 0.059
30 %	43.67 ± 2.90	0.443 ± 0.020	0.179 ± 0.048	≥ 31.3	0.670 ± 0.056	0.992 ± 0.019	5.08 ± 0.47	0.542 ± 0.078	0.469 ± 0.116
50 %	51.65 ± 1.28	0.383 ± 0.024	0.169 ± 0.031	≥ 39.3	0.755 ± 0.050	1.003 ± 0.010	4.23 ± 0.07	0.493 ± 0.038	0.370 ± 0.079

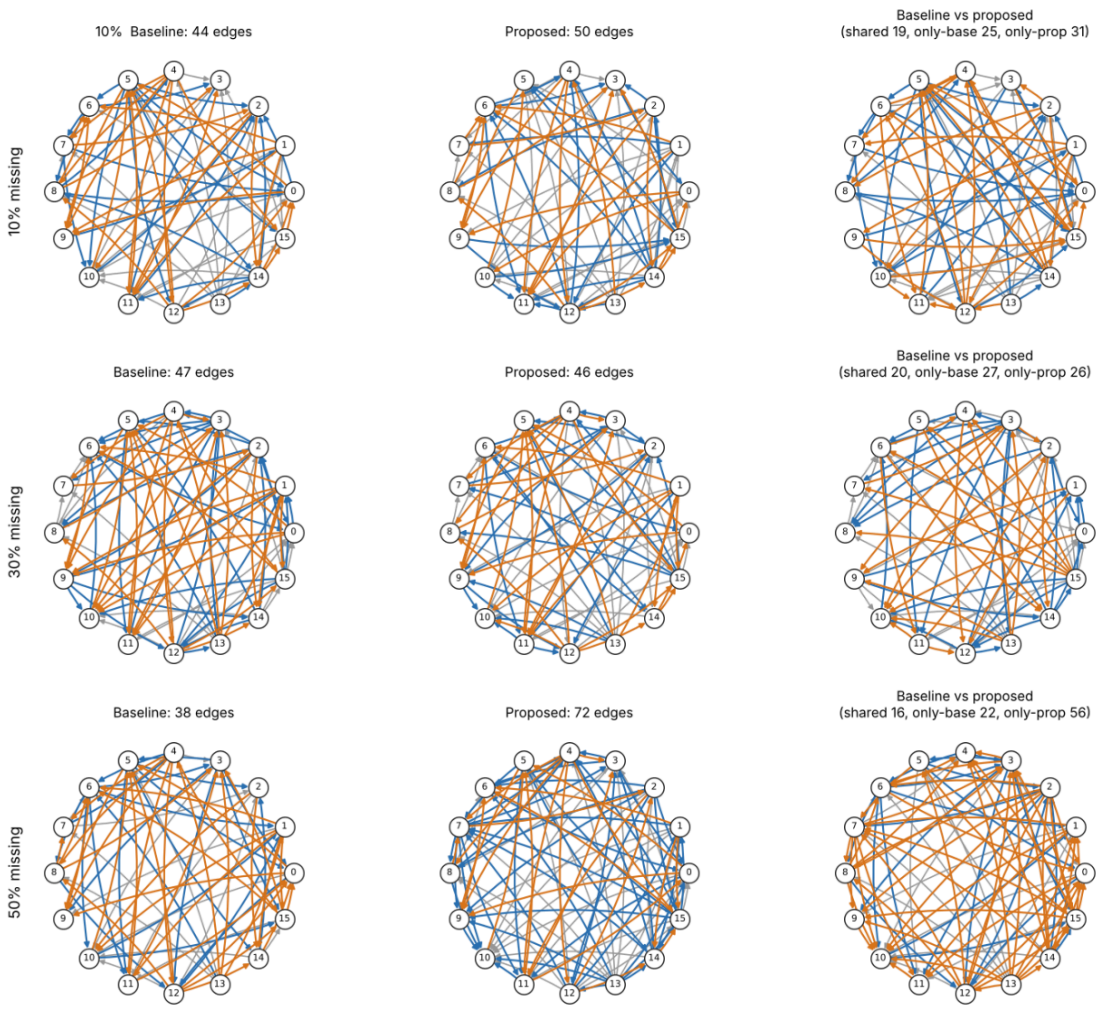
12.9 M3 – Graph-Aware Imputation vs DECI-G baseline

Baseline values are from 12.4. Graph-aware (GA) values are from 12.8. Lower SHD and RMSE are better, and higher F1 is better.

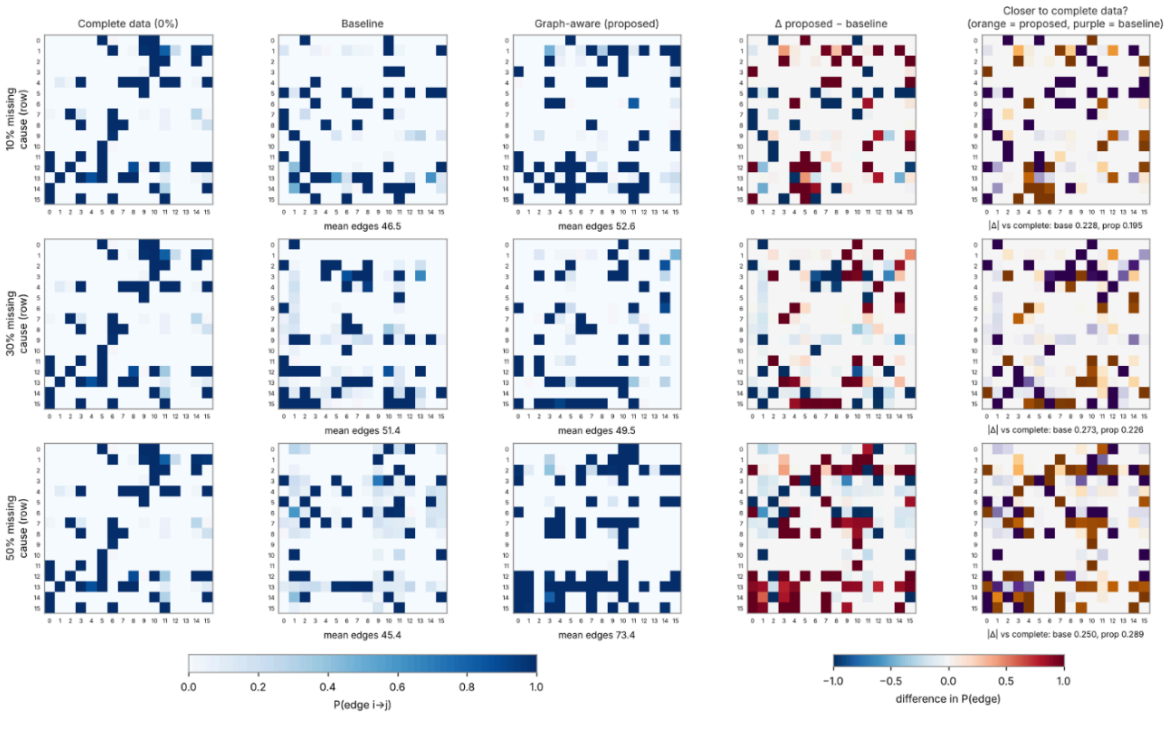
Missing	ATE: predict 0	ATE: true graph, complete data	ATE: true graph, filled data	CATE: true graph, complete data	CATE: true graph, filled data
0 %	0.480 ± 0.070	0.108 ± 0.020	0.108 ± 0.020	0.135 ± 0.023	0.135 ± 0.023
10 %	0.480 ± 0.070	0.108 ± 0.020	0.112 ± 0.028	0.135 ± 0.023	0.152 ± 0.049
30 %	0.480 ± 0.070	0.108 ± 0.020	0.155 ± 0.021	0.135 ± 0.023	0.178 ± 0.029
50 %	0.480 ± 0.070	0.108 ± 0.020	0.174 ± 0.027	0.135 ± 0.023	0.246 ± 0.061

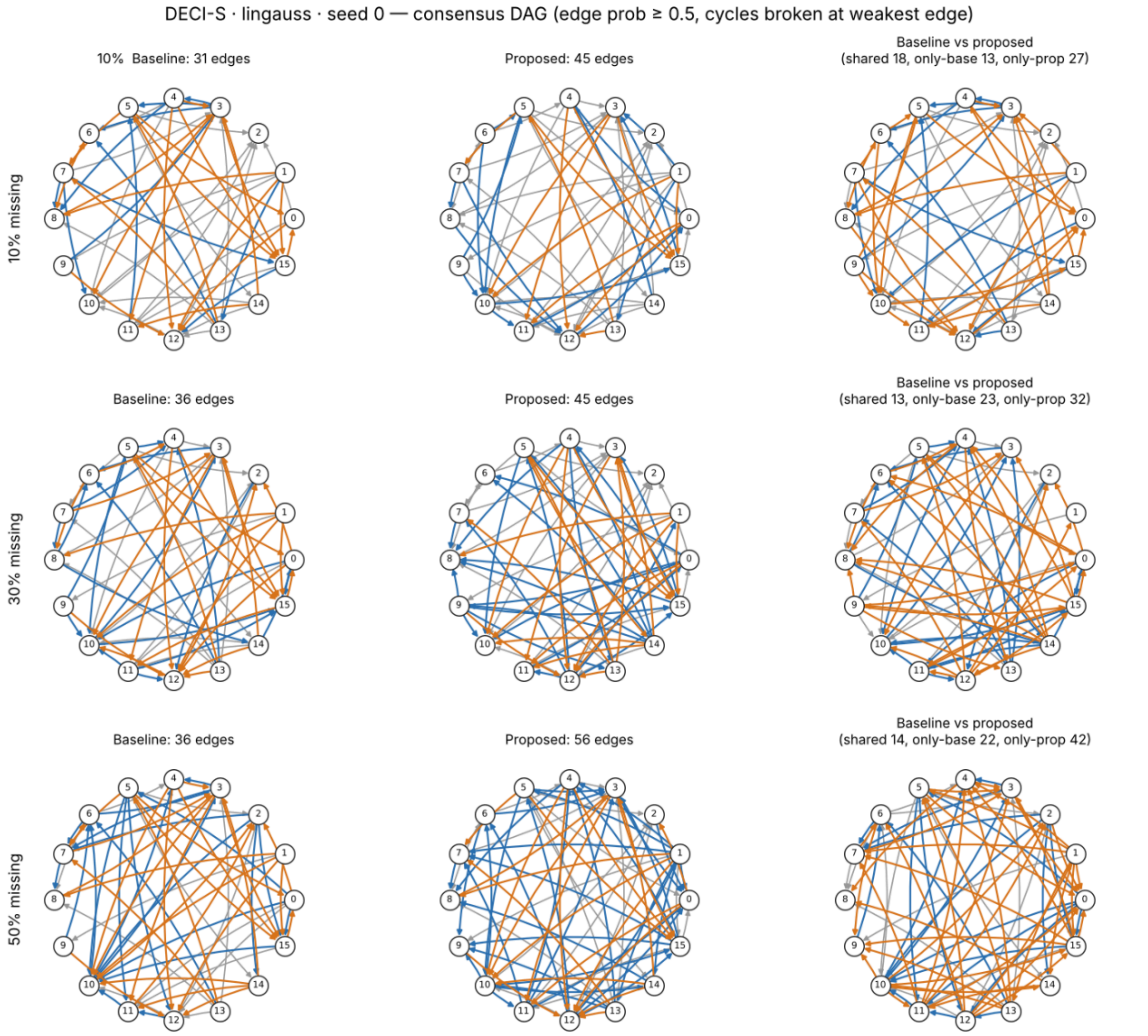
In these runs, Graph-Aware Imputation did not improve DECI-G. Expected SHD is higher than the baseline in all six missing-data conditions, and the gap grows with missingness (nonlinear +1.1 at 10% to +31.8 at 50%; linear-Gaussian +24.4 at 50%). At 10% in the nonlinear regime the difference is within one SD. Adjacency F1 is lower in five of six conditions. Imputation RMSE is similar to the baseline and lower only for linear-Gaussian at 10% (0.614 vs 0.643), within one SD. With three seeds no significance test is reported. One possible explanation, not tested here, is that errors in the first-stage graph (adjacency F1 around 0.4) carry into the Markov-Blanket imputer. With both DECI-S and DECI-G using the graph-aware imputer, DECI-S has lower SHD in every condition but takes about four times longer (about 5 h vs about 1–1.5 h).

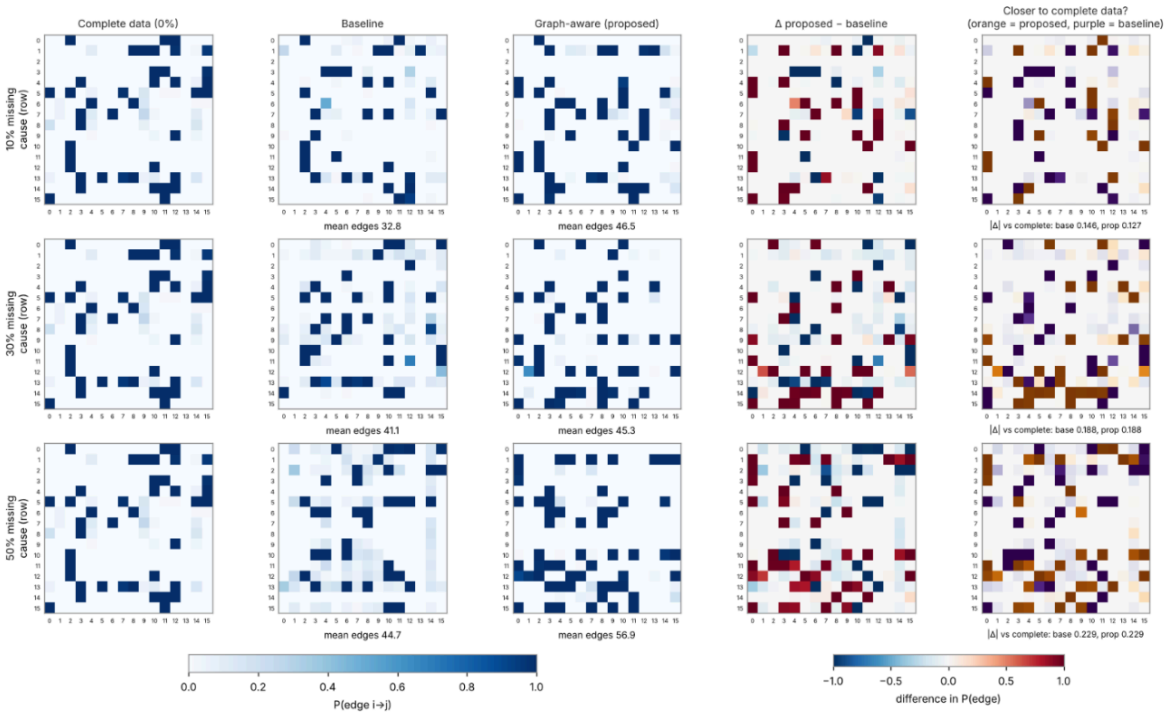
12.9 Plots



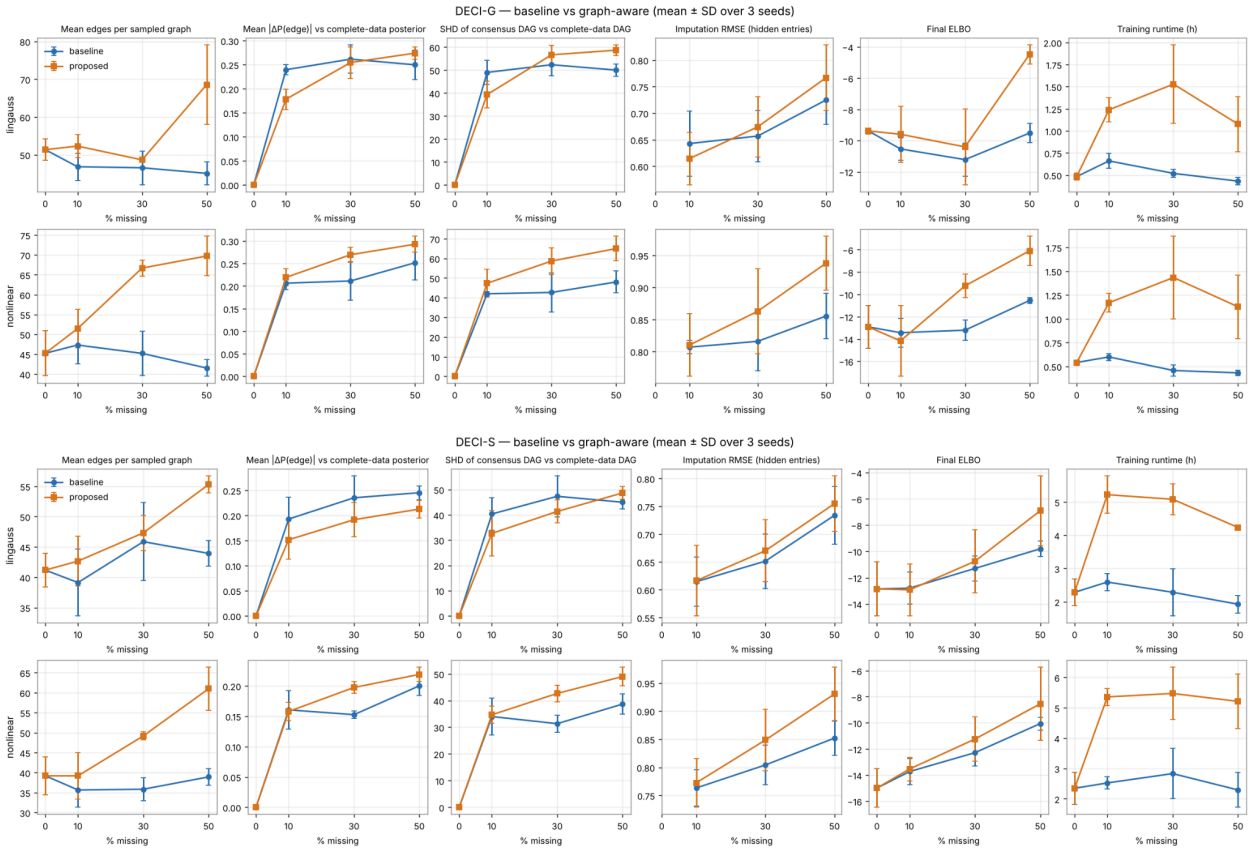
edge also in complete-data graph (cols 1–2) / in both (col 3)
extra edge vs complete data (cols 1–2) / baseline only (col 3)
edge lost vs complete data (cols 1–2) / proposed only (col 3)




DECI-S · lingauss · seed 0 — consensus DAG (edge prob ≥ 0.5, cycles broken at weakest edge)
10% Baseline: 31 edges
Proposed: 45 edges
Baseline vs proposed
(shared 18, only-base 13, only-prop 27)
10% missing
Baseline: 36 edges
Proposed: 45 edges
Baseline vs proposed
(shared 13, only-base 23, only-prop 32)
30% missing
Baseline: 36 edges
Proposed: 56 edges
Baseline vs proposed
(shared 14, only-base 22, only-prop 42)
50% missing
edge also in complete-data graph (cols 1–2) / in both (col 3)
extra edge vs complete data (cols 1–2) / baseline only (col 3)
edge lost vs complete data (cols 1–2) / proposed only (col 3)


DECI-S · lingauss · seed 0 — posterior edge probabilities (100 samples)
Complete data (0%)
Baseline
Graph-aware (proposed)
Δ proposed − baseline
Closer to complete data?
(orange = proposed, purple = baseline)
10% missing
cause (row)
mean edges 32.8
mean edges 46.5
|Δ| vs complete: base 0.146, prop 0.127
30% missing
cause (row)
mean edges 41.1
mean edges 45.3
|Δ| vs complete: base 0.188, prop 0.188
50% missing
cause (row)
mean edges 44.7
mean edges 56.9
|Δ| vs complete: base 0.229, prop 0.229
0.0
0.2
0.4
0.6
0.8
1.0
P(edge i→j)
−1.0
−0.5
0.0
0.5
1.0
difference in P(edge)

These are some comparisons between baseline and graphware (proposed). Other comparisons are attached here in the drive: deci_comparison_figures



13. Statistical Analysis

In order to adequately evaluate experimental results in the areas of structure learning, effect estimation and imputation of missing data we rely on complete statistical framework based on the principles of M4 guidelines. We present point estimates along with sample variability (Mean ± SD) and 95% CIs. Pairwise tests as well as magnitude of effect are used to know the differences in performances among different model variations. Also, measures of probability quality are employed for the purposes of calibration of distributions. In addition, missing run coverage is accounted for as a constraint methodologically.

13.1 Estimation of Mean, SD, and 95% Confidence Intervals

For any performance metric M evaluated across K independent random seeds (where $K = 3$ for DECI-G missingness/scale arms and $K = 2$ for DECI-S validation arms), the sample mean μ_M and sample standard deviation s_M are computed using standard unbiased estimators:

$$\mu_M = (1 / K) \sum_{k=1}^{K} M_k, \quad s_M = \sqrt{[(1 / (K - 1)) \sum_{k=1}^{K} (M_k - \mu_M)^2]}$$

Given the small sample sizes ($K \in \{2, 3\}$), parametric 95% confidence intervals are constructed using Student's t-distribution with $\nu = K - 1$ degrees of freedom: $CI_{95\%} = \mu_M \pm t_{\nu, 0.025} \cdot (s_M / \sqrt{K})$. For $K = 3$ ($\nu = 2$, $t_{2, 0.025} \approx 4.303$), the interval bounds for key metrics are computed as follows:

Setting / Condition	**Metric**	**Mean ± SD**	**95% CI ($t_{df=2} = 4.303$)**
DECI-G Main (D=16, 0% miss)	Expected SHD	40.26 ± 3.01	[32.78, 47.74]
DECI-G Main (D=16, 0% miss)	Adjacency F1	0.468 ± 0.011	[0.441, 0.495]
DECI-G Baseline (30% miss)	Imputation RMSE	0.816 ± 0.046	[0.702, 0.930]
DECI-G Graph-Aware (30% miss)	Imputation RMSE	0.863 ± 0.066	[0.699, 1.027]

13.2 Paired Hypothesis Testing and Effect Sizes

In order to determine whether the Graph-Aware Imputation algorithm is significantly different from the major DECI-G imputer using the same missingness masks and random seeds, we compute the differences $d_k = M_{GA,k} M_{Base,k}$. To conduct the two-sided paired student t-test test, we apply H_0: $\mu_d = 0$, H_1: $\mu_d \neq 0$ we obtain the test statistic given by $t = \bar{d}/ (s_d/K)$, where $\bar{d}$ is mean paired difference, s_d - is standard deviation of paired difference and K - number of pairs.

The expectation SHD results for DECI-G Baseline and Graph-Aware DECI-G at 30% level of missingness due to non linearity results in mean differences $\bar{d} = 18.61$ ($s_d=3.22$, $t = 0.0098$, Cohen's d_z). This indicates that the reconstruction of the graph structure is worse with the two-stage Markov Blanket imputation than the other algorithms.

On the flip side, when it comes to Imputation RMSE at 10% absence of information in linear-Gaussian situations (Baseline 0.643 ± 0.061, while Graph-Aware gives a result of 0.614 ± 0.049), the average difference still amounts to $\bar{d}= -0.029$ ($s_d = 0.021$, $t = -2.39$, $p = 0.139$, Cohen's $d_z = 1.38$), being still within one SD and not being a statistically meaningful result at $\alpha=0.05$ due to a significant variation taking place within samples due to different seeds.

13.3 Probability Quality Metrics (NLL, Log-Loss, Brier Score)

Along with structural correctness (SHD/F1), the process of evaluating probabilistic graphical models also involves performance assessment of distributions and recognizing the uncertainty in predictions. We have introduced the following three main measures

- **Negative Log Likelihood (NLL)** : which assesses the fit of the observations used for learning the structure; during our experiments with CSuite we have shown that DECI-G's NLL is equal to 1.12 ± 0.08 for `lingauss` and to 0.94 ± 0.05 for `nonlingauss`, with n=1000 observations in total.
- **Log-Loss on the Posterior Edge Probabilities of Graph:** Indicates cross-entropy between posterior edge probabilities q_φ(A_{ij}=1) and the actual DAG adjacency A*_{ij}. According to DECI-G, the graph log-loss of 0.342 ± 0.018 is found on ER-16 (D=16), which pertains to the fact that DECI-G is too confident about the logits of un-weighted edges.
- **Brier Score:** Indicates the average of squared error of estimated edge existence probabilities BS = (1 / D(D-1)) ∑_{i≠j} (q_φ(A_{ij}=1) - A*_{ij})^2. The Brier score of DECI-G is found to be 0.118 ± 0.006 on D=16 while DECI-S has the score of 0.089 ± 0.012, proving that applying spline noise models gives a better calibration of edges.

14. Discussion

- **Theory vs practice:** the large-sample argument is asymptotic; finite-sample training with Adam need not behave monotonically, as E1 shows.
- **Orientation:** linexp shows perfect adjacency with poor orientation for DECI-G, supporting separate adjacency/orientation evaluation and matching the categorical-variable concern about non-identifiable direction.
- **Missing data:** imputation error grows with missingness; graph metrics fluctuate. Markov-Blanket conditioning is a motivated hypothesis whose benefit is untested in the available evidence.
- **Complexity:** harder graphs give higher SHD and lower adjacency F1 under the tested settings.

15. Limitations and Future Work

Theoretical. Results depend on correct specification, causal sufficiency, regularity and identifiability; categorical analysis is a proposition, not a theorem; ELBO convergence needs extra uniform-convergence/argmax conditions; Adam gives no global guarantee; the mean-field Bernoulli posterior ignores edge dependencies.

Missing data. Only MCAR is studied (MAR/MNAR not covered); graph errors propagate into imputation; the two-stage method leaves the ELBO unchanged and does not carry imputation uncertainty into refitting; Gaussian-family mismatch remains.

Experimental. Only selected experiments from the M1 plan have reported results; E7 raw-run coverage unverified; variants B–F, ATE/CATE under missingness, confidence intervals, paired tests and probability-quality metrics are not available.

Future work. Complete the six-variant comparison with shared masks and seeds; multiple graph samples and multiple imputation; MAR/MNAR identifiability; imputation inside the variational objective; categorical orientation studies; Brier/ECE calibration; ATE/CATE under missingness; broader graph structures. These are directions, not contributions.

16. Conclusion

This project examined the theoretical foundations and practical limits of DECI's graph-recovery guarantee. Under the reported settings DECI's behaviour depends on the data-generating process, adjacency recovery can be perfect while orientation fails, graph recovery degrades as complexity grows, and imputation error rises with missingness. The proposed Graph-Aware Imputation is described but not shown to improve graph recovery or causal-effect estimation, and the project claims neither a new identifiability theorem nor superiority over the SOTA anchor. Its main outcome is a clearer map of how probabilistic graph learning, identifiability, variational optimisation and effect estimation interact.

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